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Formation, Electronic Structure, and Defects of Ni Substituted Spinel Cobalt Oxide: a DFT+U Study

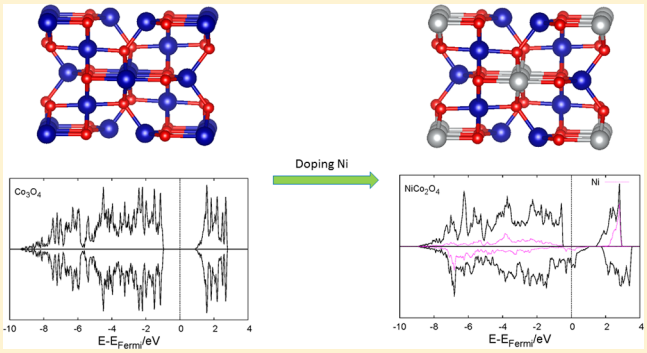
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Supporting Information

ABSTRACT: Nickel substituted spinel cobalt oxide is a promising technological material with complex electronic and magnetic structures. Understanding these structures is important for improving the material's performance in various applications. We have carried out first-principles calculations on the formation, electronic properties, and defects of bulk $NiCo_2O_4$ using density functional theory (DFT) with on-site Hubbard U terms on the transition metal d states. Analysis of the electronic structure of $Ni_xCo_{3-x}O_4$ as a function of $x = 0-1$ shows that Ni acts as a p-type dopant in Co_3O_4, gradually transforming the minority spin channel from insulating to conducting. As a result, the inverse spinel $NiCo_2O_4$ (NCO) is found to have a ferrimagnetic half-metallic ground state with fractional valence on Ni and Co cations at tetrahedral sites (Td), in agreement with experimental observations. Projected densities of states confirm that the states around the Fermi energy originate from Ni and Co(Td) 3d states hybridized with oxygen 2p orbitals. The influence of two common defects, Ni ↔ Co(Td) exchanges and oxygen vacancies, on the structural and electronic properties has been also investigated. Our results are consistent with the experimental observation that intermediate structures between inverse spinel and normal spinel occur frequently in NCO. Oxygen vacancies are predicted to occur more frequently at sites coordinated to a larger number of Ni ions and found to have only minor effects on the conductivity and magnetic structure.



INTRODUCTION

Nickel cobaltite $NiCo_2O_4$ (NCO) is an emerging technological material with a variety of promising applications, ranging from supercapacitors[1−4] to catalysts for CH_4 and low temperature CO oxidation.[5−7] For instance, a recent study has shown that NCO can completely oxidize methane to CO_2 and water in the temperature range of 350−550 °C.[7] Considering that NCO is also a cheap material made of earth-abundant elements, this result suggests that NCO may be a better methane oxidation catalyst than typical precious-metal-based catalysts.[8]

NCO is generally considered to have an inverse spinel structure with mixed valence, typically expressed as $Co_x^{2+}Co_{1-x}^{3+}[Co^{3+}Ni_{1-x}^{2+}Ni_x^{3+}]O_4$, where tetrahedral (Td) sites are occupied by both Co^{2+} and Co^{3+} ions, and octahedral (Oh) sites are occupied by Ni^{2+}, Ni^{3+}, and Co^{3+} ions.[9−11] However, evidence for Ni(Td) ions is also reported, indicating that Ni(Oh) ↔ Co(Td) exchanges can take place rather easily.[9] Co(Oh) ions are nonmagnetic, whereas Co(Td) and Ni(Oh) tend to have antiparallel spins (see Figure 1c), so that the material is usually found to be ferrimagnetic.[10,12] However, synthesis conditions can influence the magnetic order, which has sometimes led to contradictory results.[12−14] Only a few studies on NCO single crystals have been conducted;[9,15,16] these confirmed that NCO is ferrimagnetic and metallic, and the electronic and magnetic behaviors are strongly correlated with the concentration of Ni^{3+}(Oh) ions.[17] NCO was also reported to become unstable around 600 °C in vacuum and to partially decompose into NiO and $Ni_xCo_{3-x}O_4$.[9] The loss of oxygen indicates that oxygen vacancies are formed in this process. Oxygen vacancies are common and important defects in metal oxides,[18] but their effects on NCO are not clearly understood. Theoretical studies on NCO are also scarce; various bulk[13,19] and surface[7,20] properties have been investigated, but many aspects of the behavior of this material, e.g., the origin of the half-metallic ferrimagnetic structure and the influence of oxygen vacancies, are still largely unexplored.

With the growing interest in using NCO for various applications, a more detailed and complete understanding of the structure and electronic properties of this material would be desirable. This has motivated us to carry out first-principles calculations on defect-free and defected bulk of NCO using density functional theory (DFT) with on-site Coulomb repulsion (DFT+U). As a first step, we have studied the evolution from Co_3O_4 to NCO by determining the substitution

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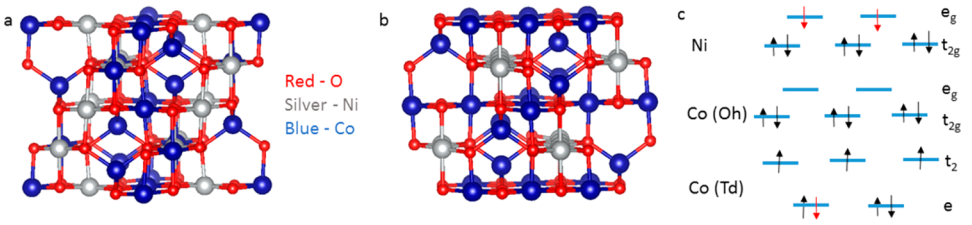


Figure 1. Conventional cell of (a) α type and (b) β type inverse spinel. The β-type inverse spinel structure was used in most calculations. (c) Spin occupations of the 3d states of Ni and Co ions in the β-type inverse spinel structure, as obtained from our calculations (section 3.2); red arrows for Ni and Co(Td) indicate fractional occupation of the corresponding orbitals.

energies and the atomic and electronic structures of $Ni_xCo_{3-x}O_4$ as a function of the doping ratio x from $x = 0$ to 1. Our results show that Ni acts as a p-type dopant in Co_3O_4 and is fractionally occupied. This leads to a half-metallic ferrimagnetic structure for NCO, consistent with previous calculations[13,19] and recent experiments.[14,17] Typical bulk defects, notably Ni(Oh) ↔ Co(Td) exchanges and oxygen vacancies, have been studied using different U values. Facile Ni ↔ Co(Td) exchange, as observed experimentally, can be reproduced using U values smaller than those determined from first-principles linear response theory. Oxygen vacancies occur preferentially at sites coordinated to a larger number of Ni ions, with a computed formation energy similar to that found for pure spinel cobalt oxide.

COMPUTATIONAL DETAILS

Spin-polarized DFT calculations were performed using the plane-wave-pseudopotential scheme as implemented in the Quantum Espresso package.[21] Exchange and correlation terms were described using the Perdew–Burke–Ernzerhof (PBE)[22] functional with on-site Coulomb repulsion U term on Co and Ni 3 d electrons. Unless otherwise specified, we used the U values determined from first-principles linear response theory,[23] denoted as U_{LR}, notably: U_{LR}(Ni) = 6.6 eV for nickel, and U_{LR}(Co) = 4.4 and 6.7 eV for Co ions at Td site and Oh sites, respectively. Ultrasoft pseudopotentials[24] were generally employed where valence electrons include O 2 s, 2 p; Co 3 d, 4 s and Ni 3 d, 4 s states. Kinetic energy cutoffs used were 50 Ryd and 500 Ryd for wave function and augmented density, respectively. Selected calculations using norm-conserving pseudopotentials[25] were also performed; in this case a kinetic energy cutoff of 100 Ryd was used. Structural optimizations were carried out by relaxing all atoms until forces were smaller than 1×10^{-3} a.u. Calculations were performed using the 56-atom conventional cubic cell containing 8 formula units (Figure 1), with a $3 \times 3 \times 3$ Monkhorst–Pack k-point mesh to sample the Brillouin zone. All results, e.g., total energies or magnetic moment, that are reported in the following will be referred to this conventional unit cell. Wannier functions were calculated using the wannier 90 code.[26]

RESULTS AND DISCUSSION

Ni Substitution in Co_3O_4: $Ni_xCo_{3-x}O_4$ (x = 0–1). We studied the formation of NCO starting from the normal spinel Co_3O_4 and successively substituting 8 Co at the octahedral sites of a 56 atom cubic supercell with Ni atoms, thus gradually increasing the doping ratio from $x = 0$ to 1 by steps $\Delta x = 1/8$. By this procedure, the resulting NCO's structure is an inverse spinel with tetrahedral sites occupied by Co only. For each x, the lattice constant was determined from the Birch-Murnagham equation of state and found to increase linearly with increasing doping ratio x, by ~0.01 Å per 1/8 change in x (see Table 1). For Co_3O_4 we determined a lattice constant of 8.149 Å, which is 0.9% larger than the experimental value 8.085 Å,[27] while the computed lattice constant for NCO is 8.237 Å, which is about 1.5% larger than the experimental value of 8.115 Å.[27]

Table 1. Lattice Constant, Bulk Modulus, and Doping Energy of $Ni_xCo_{3-x}O_4$ for Different Values of x, Calculated Using DFT+U_{LR}[a]

doping ratio x	lattice constant a_0/Å	bulk modulus B/GPa	doping energy/eV: spin parallel	doping energy/eV: spin antiparallel
0	8.149	203		
1/8	8.162	190	0.028	0.031
1/4	8.173	179	−0.009	−0.019
3/8	8.184	175	−0.056	−0.067
1/2	8.196	171	−0.062	−0.073
5/8	8.199	136	−0.086	−0.093
3/4	8.216	167	−0.095	−0.103
7/8	8.227	170	−0.113	−0.113
1	8.237	163	−0.125	−0.119

[a]The doping energies are determined for different orderings of the spins on Co(Td) ions. The preferred ordering is anti-parallel in Co_3O_4 and parallel in NCO.

Doping energies were determined from the expression

$$E_{doping} = (E_{Ni_xCo_{3-x}O_4} - E_{Co_3O_4} - x\mu_{Ni} + x\mu_{Co})/x$$

where $E_{Ni_xCo_{3-x}O_4}$ and $E_{Co_3O_4}$ are the total energies of $Ni_xCo_{3-x}O_4$ and Co_3O_4, respectively, and μ_{Co} and μ_{Ni} are the chemical potentials of Co and Ni at Oh sites. The latter were evaluated as $\mu_M = E_{MO} - \frac{1}{2}\mu_{O_2}$, where E_{MO} is the computed total energy per formula unit of the MO binary oxide (M = Co or Ni) and μ_{O_2} is the chemical potential of an oxygen molecule. As a result, E_{doping} can be rewritten as

$$E_{doping} \approx (E_{Ni_xCo_{3-x}O_4} - E_{Co_3O_4} - xE_{NiO} + xE_{CoO})/x$$

The computed doping energies for different x values are reported in Table 1. We can see that the doping energy is positive at $x = 1/8$, indicating that doping Co_3O_4 with Ni is not favorable at low concentration. For $x > 1/4$, however, the doping energy becomes increasingly negative with increasing doping ratio, indicating that further doping is favorable. For $x < 3/4$, the spins of Co ions at neighboring Td sites prefer to be antiparallel. Both antiparallel and parallel spins are present at $x = 7/8$ while the spin ordering of Co(Td) ions changes to parallel at $x = 1$ (the spins of Ni(Oh) and Co(Td) are always antiparallel). Together with the decrease of doping energies,

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Table 1 shows a small increase of the lattice constant with increasing x. The bulk modulus decreases significantly with increasing x, reaches a minimum at x = 5/8, and slightly increases afterward.

Figure 2 shows the evolution of the density of states (DOS) of $Ni_xCo_{3-x}O_4$ as a function of x. We can see that Ni doping

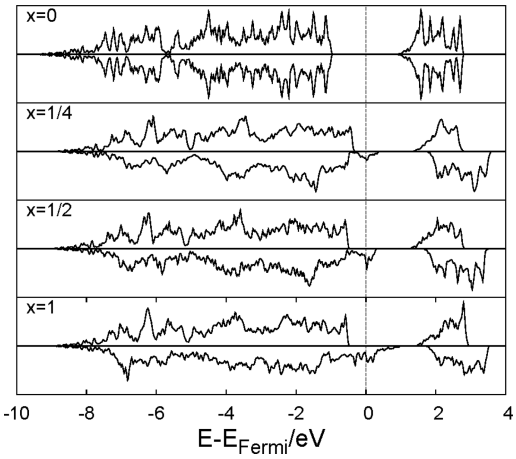


Figure 2. Electronic density of states (DOS) of $Ni_xCo_{3-x}O_4$ for different values of the doping ratio x, calculated using DFT+U_{LR}. The DOS is not affect by the spin ordering of Co(Td) ions.

gradually transforms Co_3O_4, an insulating material, into a half metal in which the majority spin channel remains insulating while the minority one becomes conducting. The valence bands for both spin channels thus shift toward the Fermi level in comparison to Co_3O_4. These results also indicate that Ni (which prefers a charge state Ni^{2+}) substituting a Co^{3+} at an Oh site acts as p-type dopant in Co_3O_4, consistent with recent experimental findings.[14,19] The presence of Ni^{2+} at an octahedral site causes indeed the formation of a $Co(Td)^{3+}$, which has only one occupied e orbital in the minority spin channel (see Figure 1). The other, empty, e orbital on Co(Td) gives then rise to a hole (acceptor) state at the top of the valence band (note that the e states of Co(Td) are part of the valence band in Co_3O_4). Similarly, a Ni^{3+} at an octahedral site (resulting in $Co(Td)^{2+}$) has an empty e_g state which also leads to a hole state at the top of or just above the valence band. Hence in both cases empty states above or at the top of the minority spin valence band are formed, indicating that Ni acts as a p-dopant in Co_3O_4.

Structure, Bonding, and Electronic Properties of NCO. We considered two different symmetries, known as α-type and β-type, of the inverse spinel crystal structure (Figure 1a-b). The β-type structure has alternating layers of -Co(Oh)-O- and —Ni-O- along the z direction in Figure 1b, whereas layers containing both Co(Oh) and Ni ions are present in the α-type structure. Total energy calculations were performed for many possible magnetic configurations with these symmetries. The β-type inverse spinel was found to be slightly more stable, by 0.05 eV per conventional cell, in comparison to the α-type structure. For both symmetries the most stable state is half-metallic and ferrimagnetic, with antiparallel spins on Ni(Oh) and Co(Td) ions and a total magnetization of 16 μ_B per conventional unit cell (Co(Oh) ions have always fully occupied t_{2g} states and no spin polarization). However, the ferromagnetic state (total magnetization of 32 μ_B) is only 0.07 eV higher in energy and also half metallic. By contrast, the ferrimagnetic configuration with insulating behavior in both spin channels has much higher energy (1.17 eV). The Densities of States of NCO's ground state and excited ferromagnetic and insulating ferromagnetic states are compared in Figure S1 of the Supporting Information.

Maximally localized Wannier functions (MLWFs)[28,29] are a useful tool for studying bonding properties; their centers provide information on the atomic oxidation states while their shapes help understand the character of the bonds. Calculation of the MLWFs for NCO's half-metallic ferrimagnetic ground state shows that there are 8, 7, and 6 d-type Wannier functions centered on the Ni, Co (Td), and Co (Oh) ions, respectively, suggesting oxidation state +3 for Co(Oh), and +2 for Ni and Co (Td) ions. However, this would introduce a charge imbalance in NCO. More detailed analysis of the band structure shows indeed that the e_g/e states of Ni and Co(Td) ions in the minority spin channel cross the Fermi energy and are not fully occupied (Figure 3a). The partial occupation of these states gives rise to both the fractional valence and the metallic behavior of the minority spin states, consistent with recent XPS data showing that both +2 and +3 oxidation states are present for Ni and Co in NCO.[12,17] As shown by the projected band structure (see the Supporting Information, Figure S2c), however, also the oxygen orbitals contribute to the states around the Fermi level, suggesting the presence of covalent interactions with the Ni and Co's e_g/e states. Due to the partial occupation of the oxygen orbitals, the fractions of Co^{2+} and Ni^{3+} states can be different and therefore NCO's composition can be better expressed as

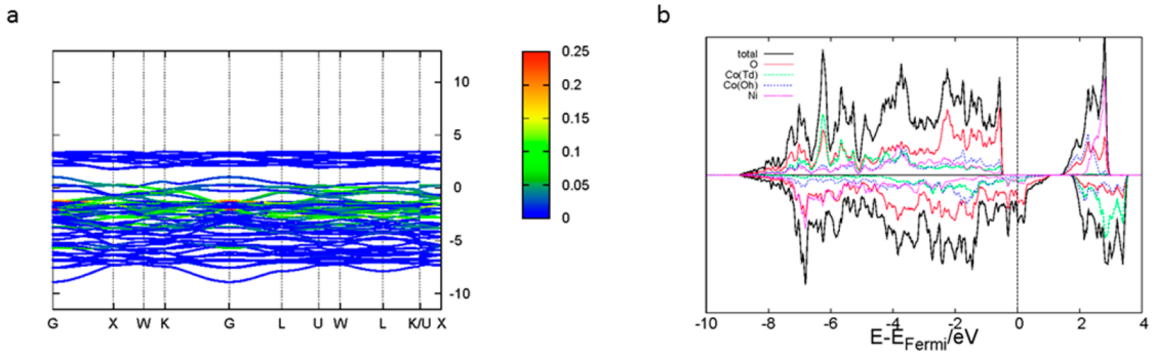


Figure 3. (a) Band structure for the minority spin states and (b) projected density of state (PDOS) of NCO, calculated using DFT+U_{LR}. The different colors in (a) reflect the contribution of the e states of Co (Td).

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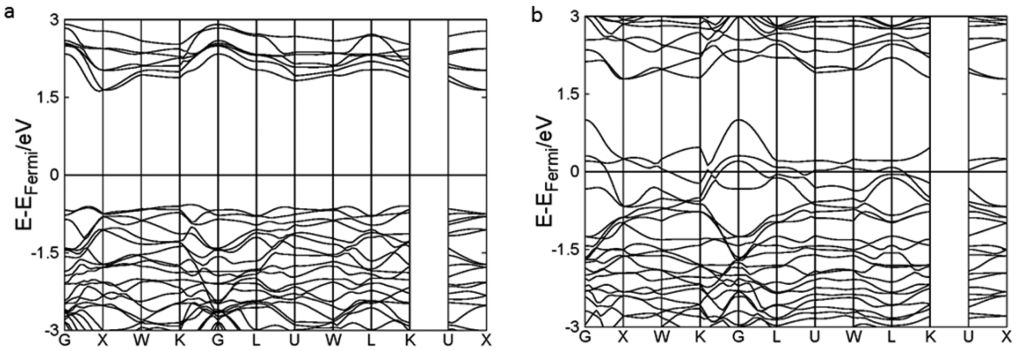


Figure 4. Band structure of NCO, calculated using DFT+U_{LR}: (a) majority and (b) minority spin states.

$Co_x^{2+}Co_{1-x}^{3+}[Co^{3+}Ni_{1-y}^{2+}Ni_y^{3+}]O_4$. For comparison, we also calculated the MLWFs of the ferrimagnetic insulating state and found that in this case one e orbital of Co(Td) is only singly occupied (see Figure 1 for comparison), consistent with a perfect $Co^{3+}[Ni^{2+}Co^{3+}]O_4$ structure, as reported in ref 9.

The projected density of states (PDOS) for NCO in its ferrimagnetic ground state is shown in Figure 3b. The main contribution to the states around the Fermi energy in the minority spin channel comes from Ni and Co(Td) as well as O ions, consistent with the presence of a significant hybridization between cation 3d and oxygen 2p orbitals. The computed atomic magnetic moments are 2.39 and −1.12 μ_B for Co(Td) and Ni respectively, in good agreement with the corresponding experimental values of 2.18 and −1.49 μ_B.[10] On the other hand, the contributions of Co(Td) and Ni to the conduction bands are very different: Co(Td) ions do not contribute to the majority spin conduction states whereas Ni ions do not contribute to the minority spin states. The detailed band structure plotted along several symmetry directions is shown in Figure 4a,b. There is an indirect band gap of about 2.0 eV along the Γ X direction for the majority spin states. However, the valence bands are quite flat so that the direct gap found near X is very similar, 2.1 eV. The minority spin bands show a stronger dispersion especially across the Fermi level.

To obtain insight into the absorption spectrum, the joint density of states (JDOS) was also calculated (see the Supporting Information, Figure S3). The norm-conserving pseudopotentials used for these calculations yield a band structure similar to the ultrasoft pseudopotential results of Figure 4a,b, with a somewhat larger bandgap of 2.4 eV for the majority spins. The shapes of the JDOS curves are similar for the majority and minority spin states, except for the longer low energy tail of the latter. This is consistent with recent experiments indicating that there is optical adsorption for NCO at relatively large wavelength (>700 nm) together with an optical band gap around 2.6 eV.[17]

Ni ↔ Co Exchanges. Motivated by the experimental evidence of Ni(Td) ions,[19] we have investigated the stability of NCO with respect to Ni(Oh) ↔ Co(Td) exchanges by calculating the formation energy (E_{sub})

$$E_{sub} = (E_{exch} - E_0)/n$$

where E_{exch} is the total energy of the crystal after exchanging Ni(Oh) and Co(Td), E_0 is the total energy of the defect free bulk in the inverse spinel structure, and n is the number of Ni sites exchanged. Results with our standard DFT+ U_{LR} setup predict however a rather large (~1.3 eV) energy cost for the Ni ↔ Co(Td) exchange, which seems at variance with experiment.[10,19] Since it is known that the choice of U in DFT+U calculations is not unique and different U values may perform better in different situations,[30,31] we then decided to examine how the formation energy E_{sub} is affected by the choice of U. Results for different n's and different choices of the Hubbard U terms are reported in Table 2. Specifically, we compare E_{sub} computed using our standard linear response U_{LR} values, with results obtained using (i) pure PBE (U = 0) and (ii) U values (U_{eff}) suggested by previous studies, i.e., U_{eff} = 5.5 and 3.0 eV for Ni[32] and both types of Co ions,[33,30] respectively. As a reference, the DOS of NCO calculated with these different choices of U are shown in the Supporting Information (Figure S4). We notice that all U values predict the material to be ferrimagnetic with total magnetization of 16 μ_B per conventional unit cell and the minority spin states to be conducting. The shapes of the DOS obtained with different U values are also similar, but pure PBE shows no gap for the minority spin states whereas a gap occurs in both DFT+U_{LR} and DFT+U_{eff} calculations.

Table 2. Computed Ni(Oh) ↔ Co(Td) Exchange Energies, E_{sub}, and Formation Temperatures, T_{sub} (in Parentheses), for Different Exchange Ratios R and Different U Values[a]

	E_{sub}/eV (T_{sub}/K)		
method	R = 1/8	R = 1/4	R = 1
PBE	0.157 (438)	0.151 (672)	0.113
DFT+U_{eff}	0.044 (123)	0.042 (185)	0.062
DFT+U_{LR}	1.305 (3541)	1.288 (5887)	1.314

[a]For example, R = 1/8 corresponds to one Co(Td) ↔ Ni exchange per conventional (56 atoms) cell.

As shown in Table 2, our computed substitution energies do not change monotonically with increasing U value. With PBE and intermediate U values (U_{eff}), the substitution energies are very small, which seems to provide the best agreement with the experimental observation that intermediate structures between inverse spinel and normal spinel occur frequently.[10,19] By contrast, the Ni ↔ Co(Td) exchange is energetically very costly with large U values (U_{LR}), indicating that U_{LR} does not predict well the thermodynamic properties. To make closer connection with experiment, we also determined the formation temperature $T_{sub} = \Delta E_{sub}/\Delta S$, where ΔE_{sub} is the formation energy difference due to the exchange, and ΔS is the corresponding change of configurational entropy (we do not consider the change of vibrational entropy, which is expected to be small for this system[34]). ΔS was determined from the expression $\Delta S = k_b \ln \frac{W_1}{W_0}$, where W_0 and W_1 are the number of

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configurations corresponding to the initial and final values of the exchange ratio. PBE and PBE+U_{eff} predict moderate exchange formation temperatures, in agreement with experimental observations,[9] whereas PBE+U_{LR} strongly overestimates the temperature (see Table 2). Increase of the substitution ratio R from 1/8 to 1/4 Ni gives slightly lower E_{sub} but higher T_{sub} due to a smaller ΔS. After the Ni ↔ Co(Td) exchange, the Ni ions at the Td sites tend to be spin parallel to the Co at the other Td sites. The total magnetization does not change and the DOS changes are also minor using both U_{eff} and U_{LR} (Figure S5). Finally, complete Ni ↔ Co(Td) exchange (R = 1 in Table 2) leads to NCO in the normal spinel structure, for which the values of E_{sub} are similar to those at smaller exchange ratios.

Oxygen Vacancies. NCO is unstable and forms NiO and $Ni_xCo_{3-x}O_4$ above 650 °C.[9] The decomposition is associated with a loss of oxygen and thus to creation of oxygen vacancies. The formation energy of oxygen vacancies is thus an important quantity for evaluating the thermal stability of NCO.

We considered 1 vacancy per conventional (56 atoms) unit cell and determined the formation energy from the expression

$$E_{form} = E_{def} + \frac{1}{2}\mu_{O_2} - E_0 - k_b T \ln W$$

where E_{def} and E_0 are the total energies of the defected and pristine crystal respectively, μ_{O_2} is the chemical potential of O_2 and $k_b \ln W$ is the configurational entropy. Two inequivalent sites for oxygen vacancies exist in NCO: site A, which has 3 Co and 1 Ni neighbors, and site B, with 2 Co and 2 Ni neighbors; to test the possible dependence of their relative stabilities on the U term, calculations for different choices of U, as described in section III.C, were performed. Results are reported in Table 3 for two different conditions, namely (a) T = 0 K and standard oxygen pressure, and (b) T = 1000 K and 10^{-9} atm oxygen pressure (decomposition conditions). This shows that the O-vacancy formation energy at site B is somewhat smaller than that at site A (independent of U), both being however similar to the O-vacancy formation energy in Co_3O_4. Moreover, both PBE+U_{LR} and PBE+U_{eff} predict negative O-vacancy formation energy for case (b), indicating that oxygen vacancies form spontaneously under such conditions. This is consistent with experiments, which indicate thermal decomposition of NCO under similar conditions (900−1000 K in vacuum).[19]

Table 3. Oxygen Vacancy Formation Energies (E_{form}) for NCO Calculated Using DFT+U with Various Choices of U[a]

material	U (eV)	lattice constant/Å	E_{form} (eV) site A	E_{form} (eV) site B
NCO	0	8.141	3.17 (0.76)	3.02 (0.61)
	U_{eff}	8.209	2.35 (−0.06)	2.26 (−0.17)
	U_{LR}	8.237	1.80 (−0.61)	1.63 (−0.77)
Co_3O_4	0	8.088	3.34 (0.93)	
	U_{LR}	8.149	1.59 (−0.82)	

[a]See text. Two situations were considered: (a) T = 0 K and standard oxygen pressure, and (b) T = 1000 K and 10^{-9} atm oxygen pressure (shown in bracket). For comparison, the O-vacancy formation energies in Co_3O_4 are also shown. For each U, the corresponding optimized lattice constant was used. For Co_3O_4, we were not able to obtain a well converged ground state for the defected crystal using U_{eff}, and therefore the corresponding value of E_{form} is not reported.

Oxygen vacancies are known to have an important influence on the electrical properties of metal oxides, often causing the material to become a n-type semiconductor.[18] It is thus interesting to study how this defect affects the structural and electronic properties of NCO. The atomic structure of an O-vacancy at site B is displayed in Figure 5b. Analysis of the atomic relaxations around the vacancy shows a large displacement of 0.26 Å away from the vacancy site for the Co(Td) bonding with the oxygen that was removed (Co1 in Figure 5b). On the other hand, the displacements of the neighboring Co(Oh), Ni, and O atoms are moderate, around 0.05 Å, with O1, O2, and O3 moving toward the vacancy site. As for the electronic structure, the formation of an oxygen vacancy gives rise to several low lying states with different total magnetizations, making the identification of the ground state quite delicate. The electronic Densities of States for defect free and defected NCO with A and B-type oxygen vacancies are compared in Figure 5a. For the majority spin states, both vacancy types result in the formation of additional states in the band gap. For the minority spin states, the states at the Fermi level split in the case of an A-type vacancy, leading to the opening of a small band gap of ∼0.3 eV; here, the low energy empty states originate from Ni ions, which do not contribute to the conduction bands of defect free bulk (Figure S6). In the case of a B-type vacancy, the minority spin states remain conducting and the overall changes in the electronic structure are minor. Since B-type vacancies are favored over A-type ones, this suggests that, at variance with other oxides, oxygen vacancies have only a minor influence on the conductivity of NCO. The occupation of the d orbitals on the Co(Td), Co(Oh), and Ni ions close to the vacancy are illustrated in Figure 5c. Interestingly, although Co(Td) has a large displacement, its occupation remains unchanged. A change occurs only for the Co(Oh) ion which becomes 2+ and assumes an antiparallel spin with respect to Ni (we note that the same results are obtained with DFT+U_{LR} and DFT+U_{eff}). Moreover, the total magnetization remains unchanged in the presence of the vacancy, possibly because the hybridization between metal and oxygen states leads to compensation of the spin polarization.

CONCLUSIONS

In this work we have used DFT+U calculations to investigate the electronic and thermodynamic properties of spinel $NiCo_2O_4$ (NCO) and to obtain insight into the origin of the high conductivity and ferrimagnetic properties of this material. We have studied the formation of NCO from Co_3O_4 focusing on the electronic structure of $Ni_xCo_{3-x}O_4$ as a function of the doping ratio x. Our results show that Ni acts a p-type dopant in Co_3O_4 and gradually makes the minority spin channel metallic. Of the two possible symmetries, α-type and β-type, of inverse spinels, NCO favors the latter, but the energy difference between the two structures is quite small. Investigation of different possible magnetic structures indicates that NCO has a ferrimagnetic ground state with a low lying ferromagnetic excited state, which may explain why different conditions of synthesis often cause major differences in the observed magnetic properties.[7] Further analysis of the ferrimagnetic ground state indicates that the e_g/e states of Ni and Co(Td) are partially occupied, resulting in fractional valence and metallic behavior, consistent with recent experimental results.[17]

We also studied the influence of two frequently observed defects, Ni ↔ Co(Td) exchanges and oxygen vacancies, on the structural and electronic properties. The computed energy cost of Ni ↔ Co(Td) exchanges is largely independent of the value

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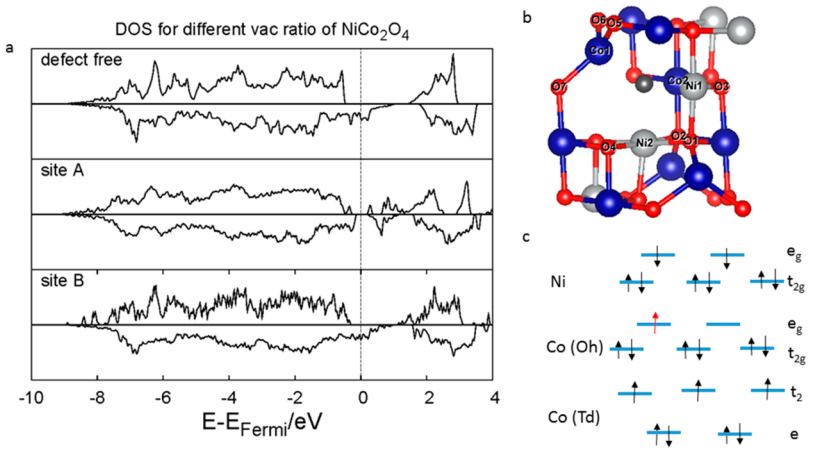


Figure 5. (a) DOS of defect free (top) and reduced NCO containing an oxygen vacancy, either at site A (middle) or at site B (bottom). (b) Atomic structure of a B-type oxygen vacancy; the dark gray ball is the vacancy site. (c) Occupation of Ni and Co d orbitals in the presence of an oxygen vacancy; the red arrow indicates a newly occupied orbital in comparison to those in defect free NCO.

of the exchange ratio, consistent with the fact that significant cation disorder is usually observed in spinel oxides.[19] Oxygen vacancies are predicted to occur more frequently at sites coordinated to a larger number of Ni ions and to form spontaneously under conditions of high temperature and low oxygen pressure, where thermal decomposition of NCO is actually observed to take place. On the other hand, oxygen vacancies are found to have only a minor effect on the magnetic and electronic properties and, therefore, do not modify the p-type character of the conductivity.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.6b03096.

(1) DOS for different magnetic structures. (2) Projected DOS for Co, Ni, and band structure with contribution from O calculated from Wannier functions. (3) Electronic structure and JDOS calculated using norm-conversing pseudopotentials. (4) Comparison of DOS calculations with different U values. (5) DOS of Ni ↔ Co exchange with different U values. (6) Projected DOS for oxygen vacancies. (PDF)

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Notes

The authors declare no competing financial interest.

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